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WHAT COMES NEXT?

Federated AI Data Lake
CHAIN++

Sovereign data for the AI economy

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ICT CLUSTER VARNA

BSMEPA
BULGARIAN SMALL AND MEDIUM ENTERPRISES PROMOTION AGENCY

ВАРНА

BRAIN++ AI FACTORY

ARC FUND

BRAIN++
AI FACTORY

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FEDERATED AI DATA LAKE

Transforming cross-border data collaboration with privacy-first architecture

EuroHPC
Joint Undertaking

Co-funded by
the European Union

WHAT COMES NEXT?

Who remembers this?

EVERYONE IN THIS ROOM ALREADY USES AI. THAT IS NOT THE QUESTION.

Do you trust AI?

In your personal life?

At work?

How much?

Why?

Rhetorical questions.

TRUST IT OR NOT, WANT IT OR NOT

You already use AI.
AI already uses you.

Not just uses — consumes.

More and more → Everywhere, often → **Soon: constantly**

Give trust a foundation — and put it to work

NOT

~~Faith~~

"I believe it's fine."

NOT

~~Hope~~

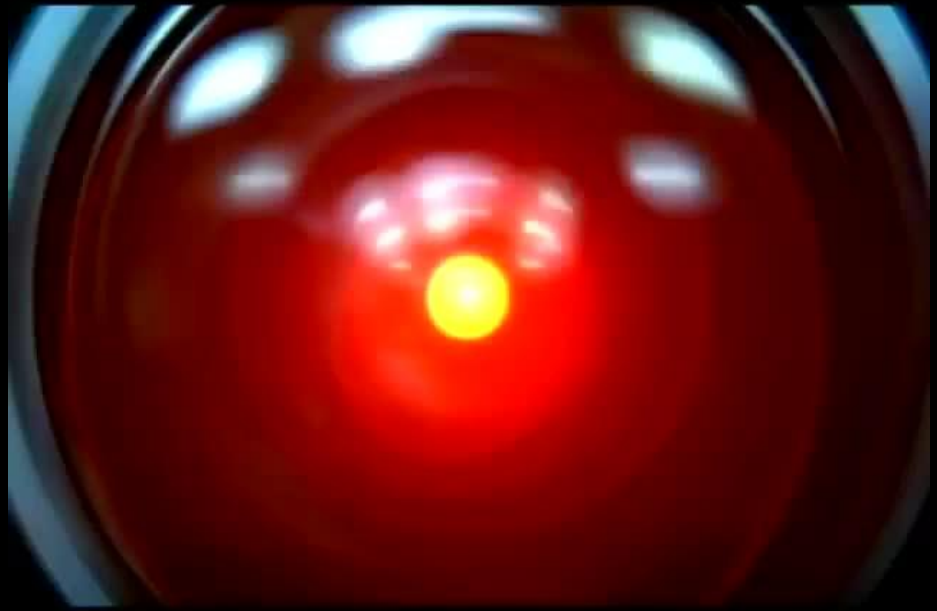
"I hope it's fine." — even more dangerous

YES

Technology

Trust you can verify — so AI consumes our data correctly, and we stay protected

Nothing dramatic. **We are building a technology of trust for the use of AI.**



**“I'm sorry, Dave.
I'm afraid I can't do that.”**

HAL 9000 — 2001: A Space Odyssey (1968)

In 2026 our data says the same to our AI:

“I'm sorry. I can't share that.”

WHAT COMES NEXT?

“I'm sorry, HAL.”

Trust — not by faith, but by design.

CHAIN++
is the answer.



For data owners, sharing is expensive — and frightening



1 Compliance paralysis
GDPR[1], NIS2[2] and sector rules turn every exchange into a legal maze — months of delay

2 Zero-trust reality
Sharing means losing control: no visibility into who accesses what, when or why

3 Technical debt
Legacy systems were never built for secure, granular, auditable external access

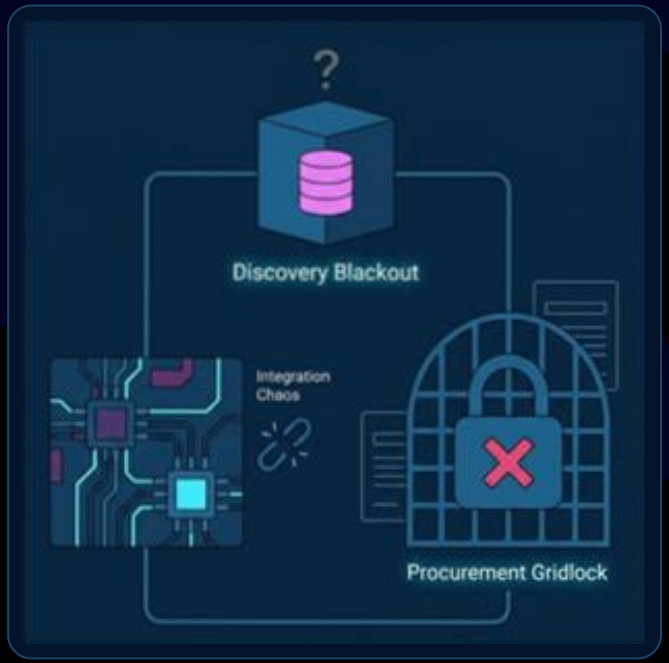
[1] GDPR — EU privacy law on collecting, using and sharing personal data [2] NIS2 — EU cybersecurity directive raising security and reporting duties

For AI builders, the data exists — but stays out of reach

1. **Discovery blackout**
   No central catalogue — valuable datasets stay invisible across organisational boundaries

2. **Integration chaos**
   Every provider has different APIs, formats and authentication — weeks of custom work per source

3. **Procurement gridlock**
   Legal negotiations, SLAs and compliance reviews turn a simple request into a 6-month project



Stop moving data to compute. Move compute to the data.

OLD MODEL

Copy, centralise, lose control

- Centralised warehouse
- Single point of failure
- Privacy & compliance risks
- Data exposure concerns
- Vendor lock-in, limited portability

NEW MODEL

Connect, federate, retain sovereignty

- Interoperable, connected network
- Secure federated nodes
- Encrypted, auditable sharing
- In-place data control
- Data remains local

Four challenges — solve them together, or none is solved



PROBLEM
Data exposure and unauthorised access



PROBLEM
Untracked interactions and compliance gaps



PROBLEM
Manual compliance burden and regulatory friction



PROBLEM
Hidden datasets and fragmented catalogues

Together they decide whether data exchange is secure, transparent and compliant in a federated environment.

One platform, four answers — data never leaves your infrastructure

SOLVES TRUST & PRIVACY



Data Sovereignty

Full ownership and control — data stays on your servers

SOLVES AUDITABILITY



Smart Contracts

Automated policy enforcement of predefined access rules

SOLVES DATA COMPLIANCE



Data Sanitization

AI-driven masking of personal data before federated exposure

SOLVES DATA DISCOVERY



Unified Discovery

One catalogue across all sources, with granular permissions

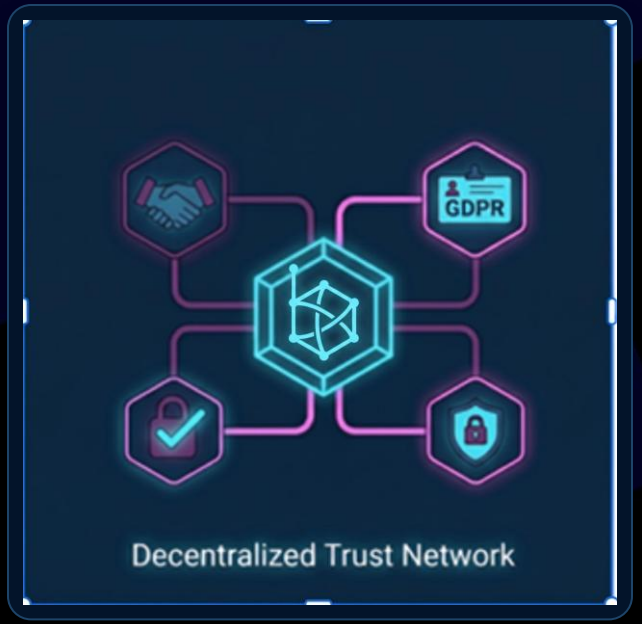
Trust is not assumed — it is cryptographically engineered

Distributed Ledger Technology[1] on Hyperledger Besu

The enterprise blockchain platform trusted by financial institutions and governments

- Peer-to-peer data sharing — no intermediaries
- Cryptographic proof of data integrity
- Privacy by design — only authorised parties see contracts
- GDPR-compliant identity management

Each node keeps its own state: no global visibility, no central point of failure



[1] DLT — a ledger shared and synchronised across many nodes, updated securely by cryptography and consensus, without a central authority

Every action leaves an immutable, provable trail



Every action — from access requests to consent updates — is recorded in a cryptographically secure, immutable audit trail.

1. **Immutable logs**
   Once written, an entry cannot be altered or deleted

2. **Verifiable transparency**
   Irrefutable proof of lineage and access — for GDPR, NIS2 and dispute resolution

3. **Enterprise-grade security**
   Enterprise consensus and secure transaction management — auditability at scale

AI agents turn legal text into code — and protect data before it moves



Policy Automation Agent

Translates GDPR clauses, DPAs and regulatory requirements into executable smart-contract logic — no manual policy coding, no legal ambiguity



Privacy-Preserving Agent

Applies anonymisation, pseudonymisation and differential privacy automatically, based on data sensitivity and access permissions

ONBOARDING

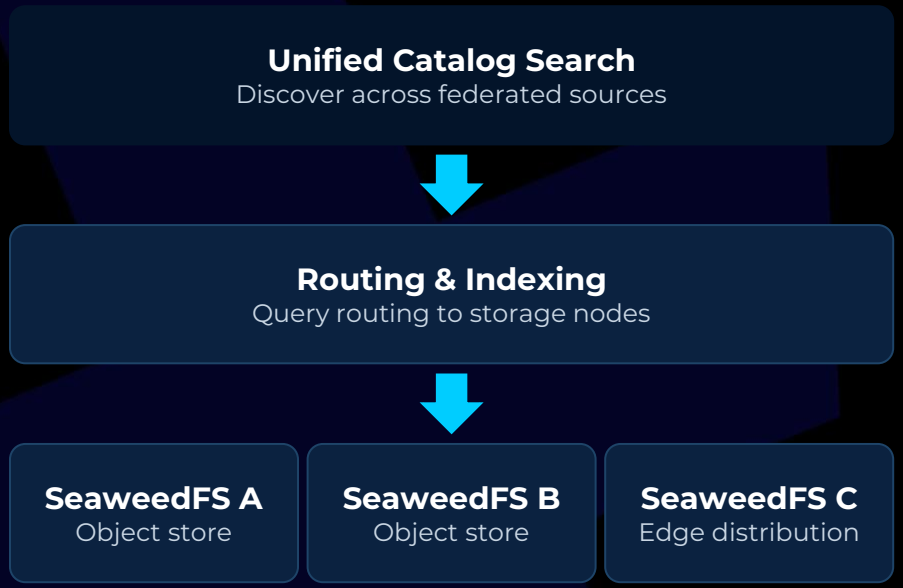
months

↓

days

Compliance workflows that needed legal teams and custom development — automated

Find any dataset across the federation — without moving it



01 Unified catalog
Search across federated sources with standardised metadata

02 Edge distribution
Data replicated geographically for low-latency access

03 Query routing
Intelligent query planning across heterogeneous sources

CHAIN++ builds on open-source, distributed object storage optimised for large-scale data lakes.

Not another walled garden — built to the European standard



Native integration with Europe's data-sovereignty initiatives ensures seamless interoperability and regulatory alignment.

Eclipse Dataspace

Open-source connector framework for trusted data exchange across organisational boundaries

Gaia-X Federation

Aligned with European cloud standards for digital sovereignty and data portability

SIMPL Ecosystem

Integrated with the EU data-spaces initiative for secure multi-party collaboration in the public sector

What Comes Next?
Beyond a data lake: a tokenized data economy



1. **Monetize idle compute**
   Turning unused resources into value

2. **Unlock locked data**
   Making dormant datasets productive

3. **Fair market incentives**
   Rewarding quality contributions — from people and AI agents

Oncology AI research: from 18 months to days

Critical medical insights unlocked — quickly and compliantly — accelerating life-saving research.

1

ONCOLOGY DATA PROVIDER

Hospital A

Automates consent, anonymisation and policy enforcement via smart contracts

2

CHAIN++

Sovereign Data Platform

Secure, auditable, privacy-preserving access to oncology datasets

3

AI RESEARCH PARTNER

Research Institute B

Accesses curated, compliant data for cancer-treatment discovery

18 months

Typical IT + legal setup today

days

With CHAIN++

Faster breakthroughs, better patient outcomes and regulatory compliance — without ever centralising sensitive data.

Funded, designed — ready to build with you

Advanced technology and strong European integration: a secure, compliant environment ready for unprecedented data collaboration.

1

Unwavering security

Built on Besu and privacy-by-design principles for data integrity and trust

2

European endorsement

Backed by competitive EU funding and aligned with Gaia-X and Simpl

3

Ready for builders

Integrate your data and services into a growing network of opportunities

Where we are today: architecture designed · EU funding secured · core team and founding consortium forming · first prototype next

Shape the standard — and stop paying the compliance tax

✓ **Faster innovation**
Finally unlock the siloed data your teams need to innovate

✓ **Zero vendor lock-in**
Enterprise-grade, privacy-first infrastructure on open European standards

✓ **Reduced compliance overhead**
Significantly less internal effort spent on legal and compliance work



Join early → a seat at the table that shapes the future standard of European data sharing.

Join us as a founding design partner

Build the future of sovereign data exchange with us. Three ways to engage:

01

Participate in the pilot

Influence platform development and validate use cases in our early-access programme

02

Become a data provider

Unlock new revenue streams by securely offering datasets and compute to the ecosystem

03

Co-create solutions

Build applications and services on the federated AI data lake that drive economic value

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That is what comes next

“Open the pod bay doors, HAL.”

With CHAIN++ the answer finally is: **Yes — safely, legally, auditably.**

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BRAIN++ AI FACTORY

EuroHPC Joint Undertaking

Co-funded by the European Union